Fresher Profiles Private Limited
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FACTORY SAFETY VOICE DETECTION SYSYTEM

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Problem Statement

Industrial environments operate under constant noise, restricted mobility, and time-critical risk conditions where traditional safety systems often fail to respond instantly. Workers cannot always access manual alert systems, leading to delayed emergency reporting and increased hazard severity. This project introduces a real-time, voice-driven safety system designed to detect critical commands such as “help”, “fire”, and “stop” even in high-noise environments.

Objectives

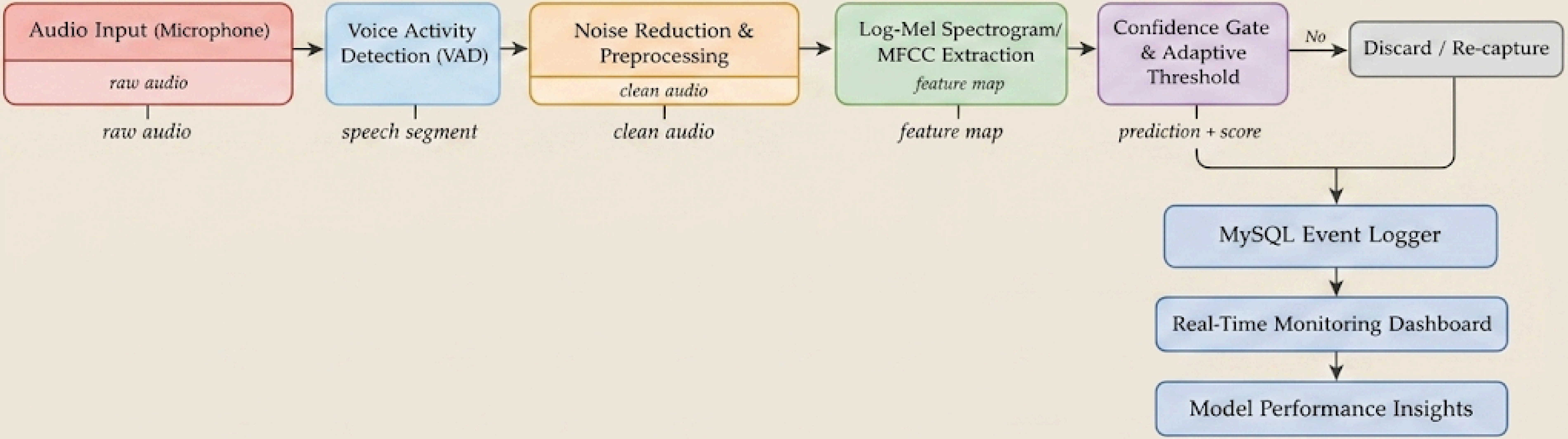
- The objective is to enable hands-free emergency reporting
- Improve response time
- Enhance industrial safety through intelligent
- context-aware decision-making.

Methodology

The system follows a real-time audio intelligence pipeline:

- *Continuous audio capture from factory environment*
- *Voice Activity Detection (VAD) isolates speech*
- *Noise reduction using spectral subtraction & Wiener filtering*
- *Feature extraction via Log-Mel spectrograms*
- *DS-CNN model performs keyword classification*
- *Confidence gating ensures reliable predictions*
- *Agentic AI layer analyzes context and triggers alerts*

System Architecture



Results & Innovation

The system demonstrates strong performance in detecting safety-critical commands even under high background noise conditions. The integration of noise reduction and Log-Mel feature extraction significantly enhances accuracy, while the DS-CNN model ensures efficient real-time inference.

The key innovation lies in the Agentic AI alert layer, which enables context-aware decision-making by analyzing multiple parameters such as confidence score, noise level, and location. This allows intelligent alert generation rather than simple detection.

Thank You